



Prepared by : Dr. Arpankumar Raval

Id: 1042

Contents

Preview.....2

Creating an Android Application.....3



Preview

- Android is an operating system and programming platform developed by Google for mobile phones and other mobile devices, such as tablets.
- It can run on many different devices from many different manufacturers. Android includes a software development kit (SDK) that helps you write original code and assemble software modules to create apps for Android users.
- The first beta version of the Android Software Development Kit (SDK) was released by Google in 2007 where as the first commercial version, Android 1.0, was released in September 2008.
- Now the Kotlin is official language for development instead Java.



Creating an Android Application

Tools required: Android Studio Version Dolphin

Table to Download links for tools:

Android Studio	https://developer.android.com/studio
---	---

Creating our first Android Application:

1. Open Android Studio.
2. In the Welcome to Android Studio dialog, click Start a new Android Studio project.
3. Select Empty Activity
4. Give your application a name such as My First App.
5. Make sure the Language is set to Kotlin.
6. Leave the defaults for the other fields.
7. Click Finish.



Setup visual studio 2022 and Web application with ASP.NET MVC 6.0

.NET
Core

In this tutorial, I'll show you how to download and install Visual Studio 2022 and construct a simple.NET Console application step by step.



Version 17.0 of Visual Studio 2022 supports applications written with .Net 6.0. We have downloaded and installed Visual Studio 2022 Community Edition Version 17.0. The Community edition of Visual Studio 2022 is available for free.

So, let's start

Step 1

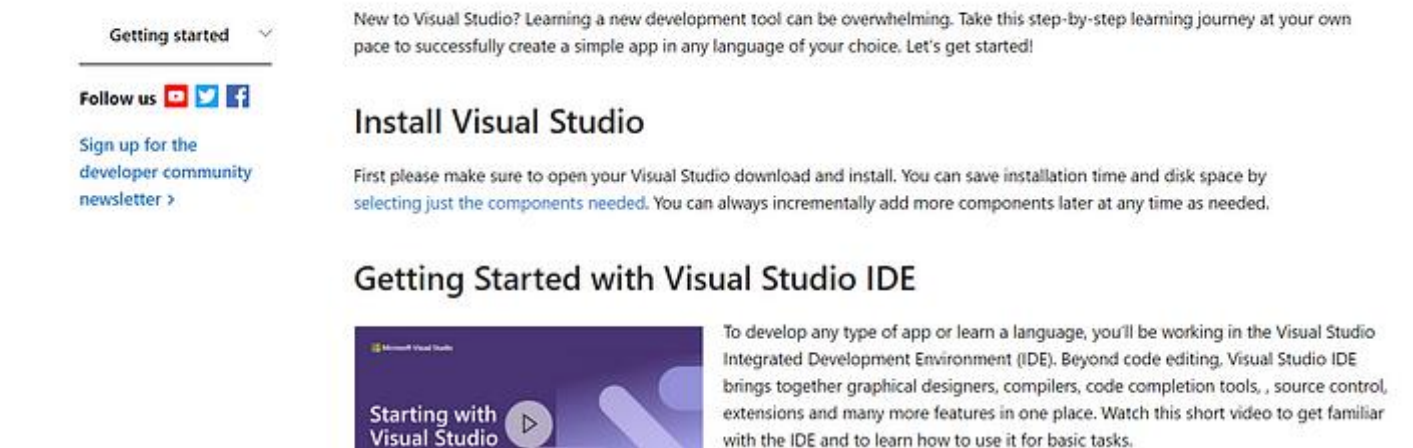
You can download Visual Studio 2022 version 17.0 from this [link](#).

Downloads

The screenshot shows the Visual Studio 2022 download page. On the left, there is a section for 'Visual Studio 2022' with the version '17.0' and a description: 'The best comprehensive IDE for .NET and C++ developers on Windows. Fully packed with a sweet array of tools and features to elevate and enhance every stage of software development.' Below this are links for 'Release notes', 'Compare Editions', and 'How to install offline'. To the right, there are four columns for different editions: 'Community' (Powerful IDE, free for students, open-source contributors, and individuals, with a 'Free download' button), 'Professional' (Professional IDE best suited to small teams, with a 'Free trial' button), 'Enterprise' (Scalable, end-to-end solution for teams of any size, with a 'Free trial' button), and 'Preview' (Get early access to latest features not yet in the main release, with 'Learn more' and 'Release notes' links).

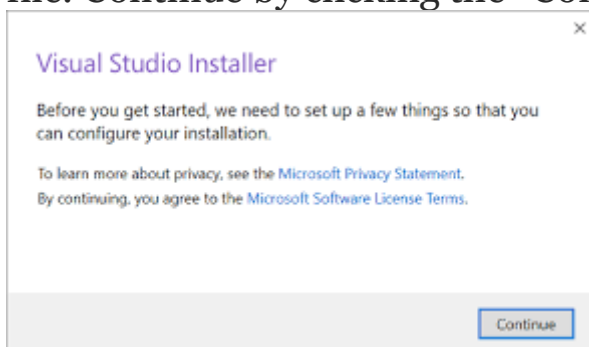
Step 2

To download the VS 2022 executable file to the downloads directory, click the “Download” button.



Step 3

The Visual Studio installer window will open when you double-click the.exe file. Continue by clicking the “Continue” button.



Step 4

After hitting the Continue button, the download and installation progress bar window will appear.

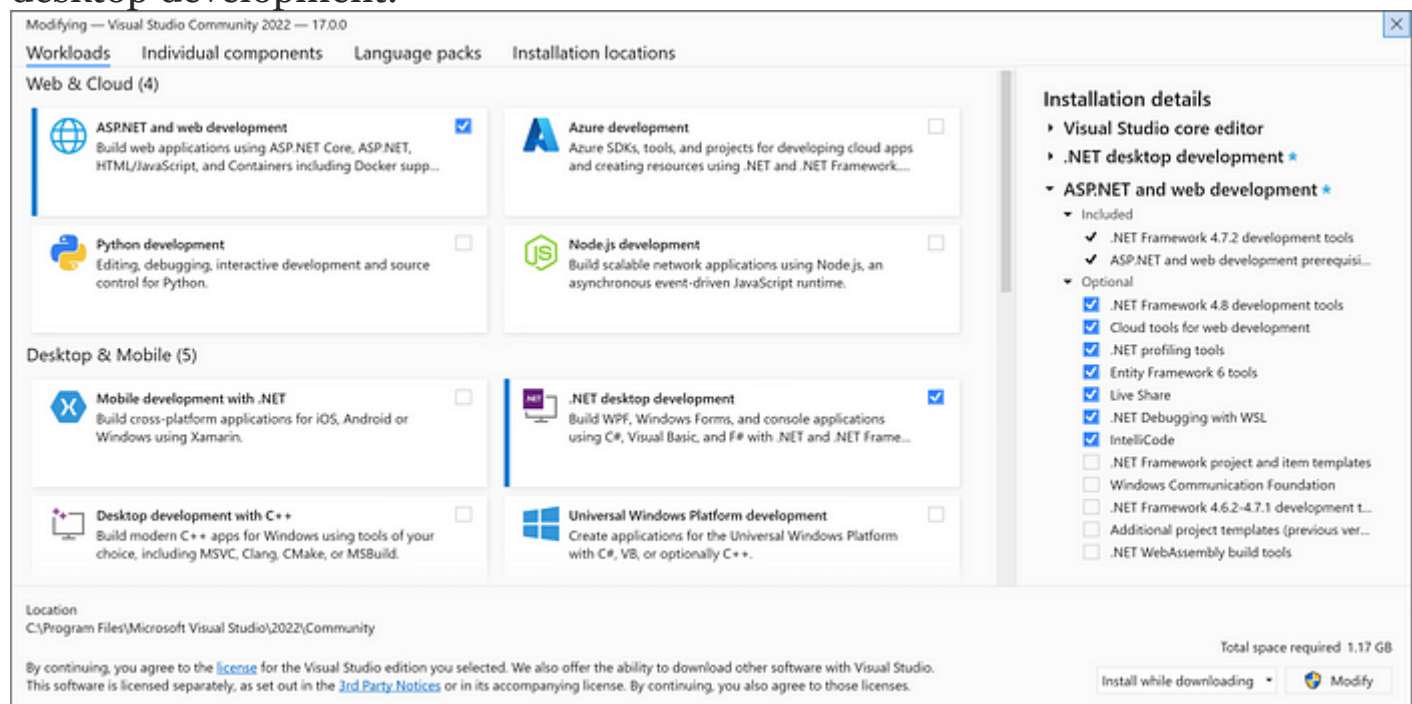
Visual Studio Installer

Almost done ... Getting everything ready.



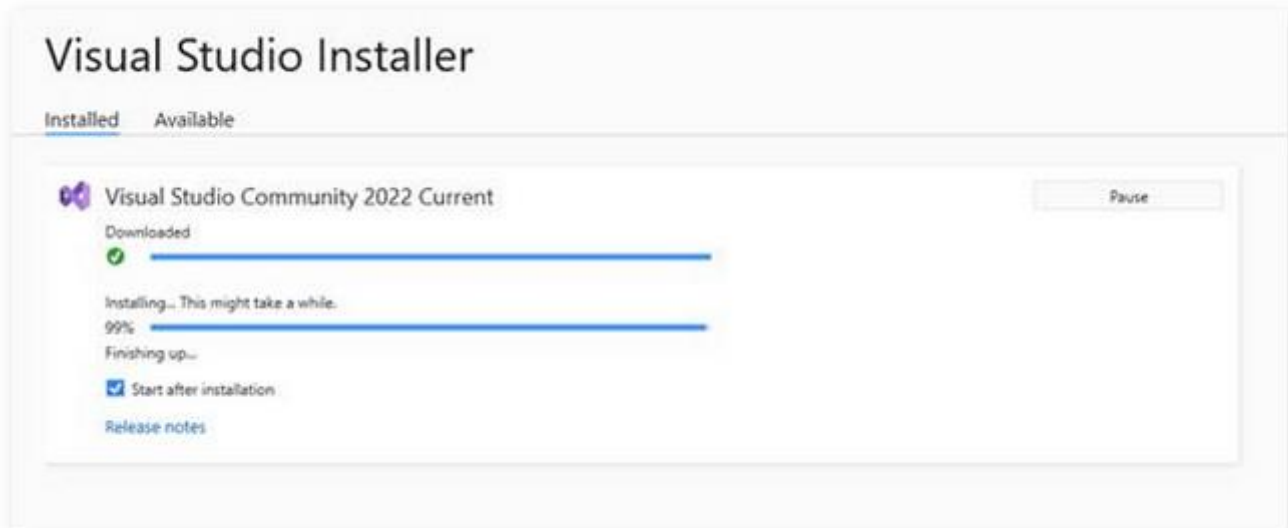
Step 5

Workloads will open after the download and installation are completed. We must decide which workloads we require. We chose ASP.NET and .NET desktop development.



Step 6

After selecting the appropriate packages, simply click the “Install” button.



When the installation is finished and we open Visual Studio for the first time, it takes 15–30 seconds for its initial internal setup.



We're getting things ready

Thanks for your patience.

• • • • •

Your existing Visual Studio profile will be loaded. It may also request that you sign in to your Microsoft account. You can establish a new Microsoft account if you don't already have one.



The sign-in process is optional, so you can skip it. But don't forget to do this! Log up with your Gmail or Microsoft account to activate Visual Studio and receive a free lifetime licence.

If you don't have an account yet, select Not now, perhaps later.

Visual Studio

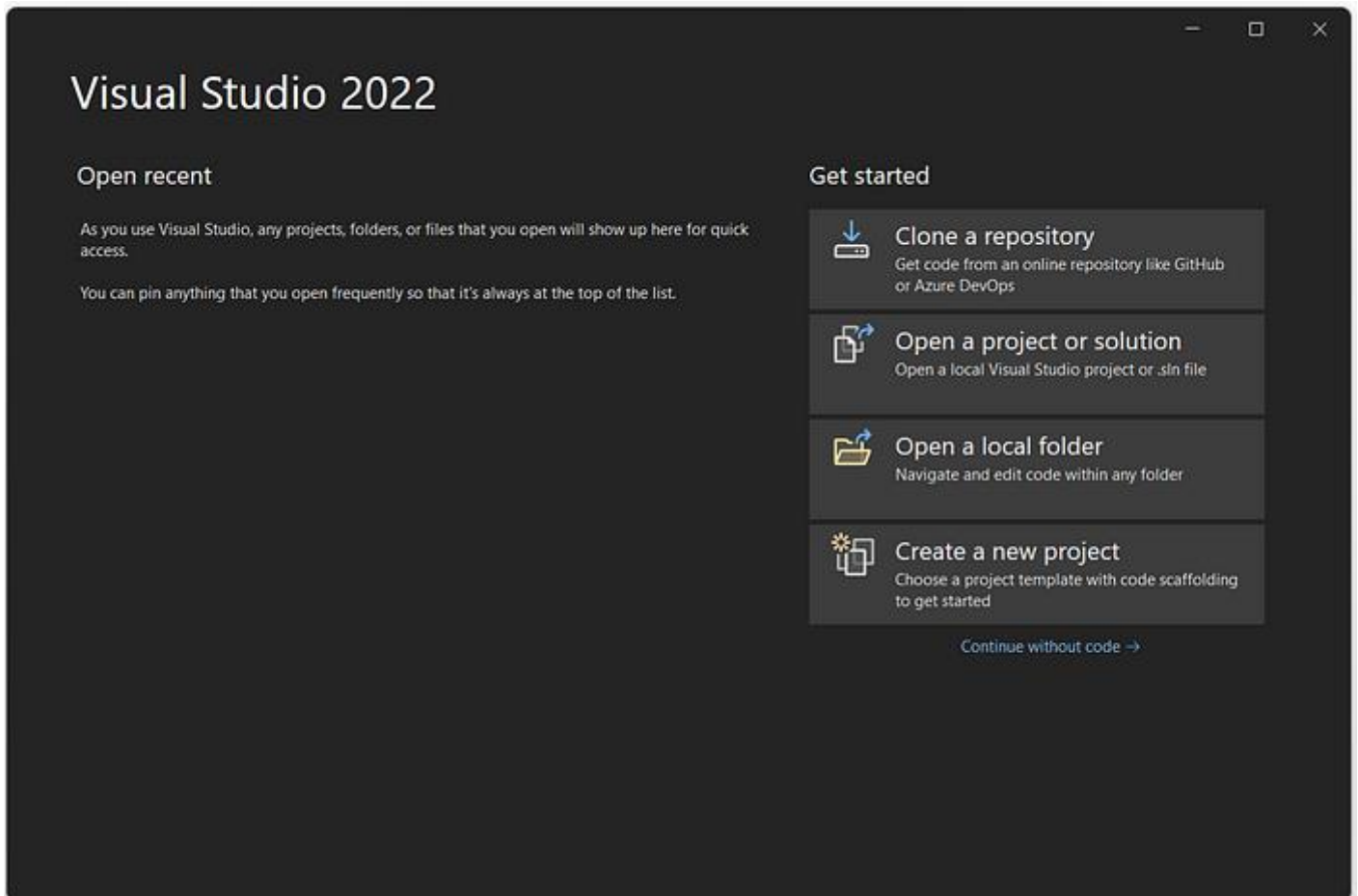
Sign in to Visual Studio

- Sync settings across devices
- Collaborate in real time with LiveShare
- Integrate seamlessly with Azure Services

Sign in

No account? [Create one!](#)

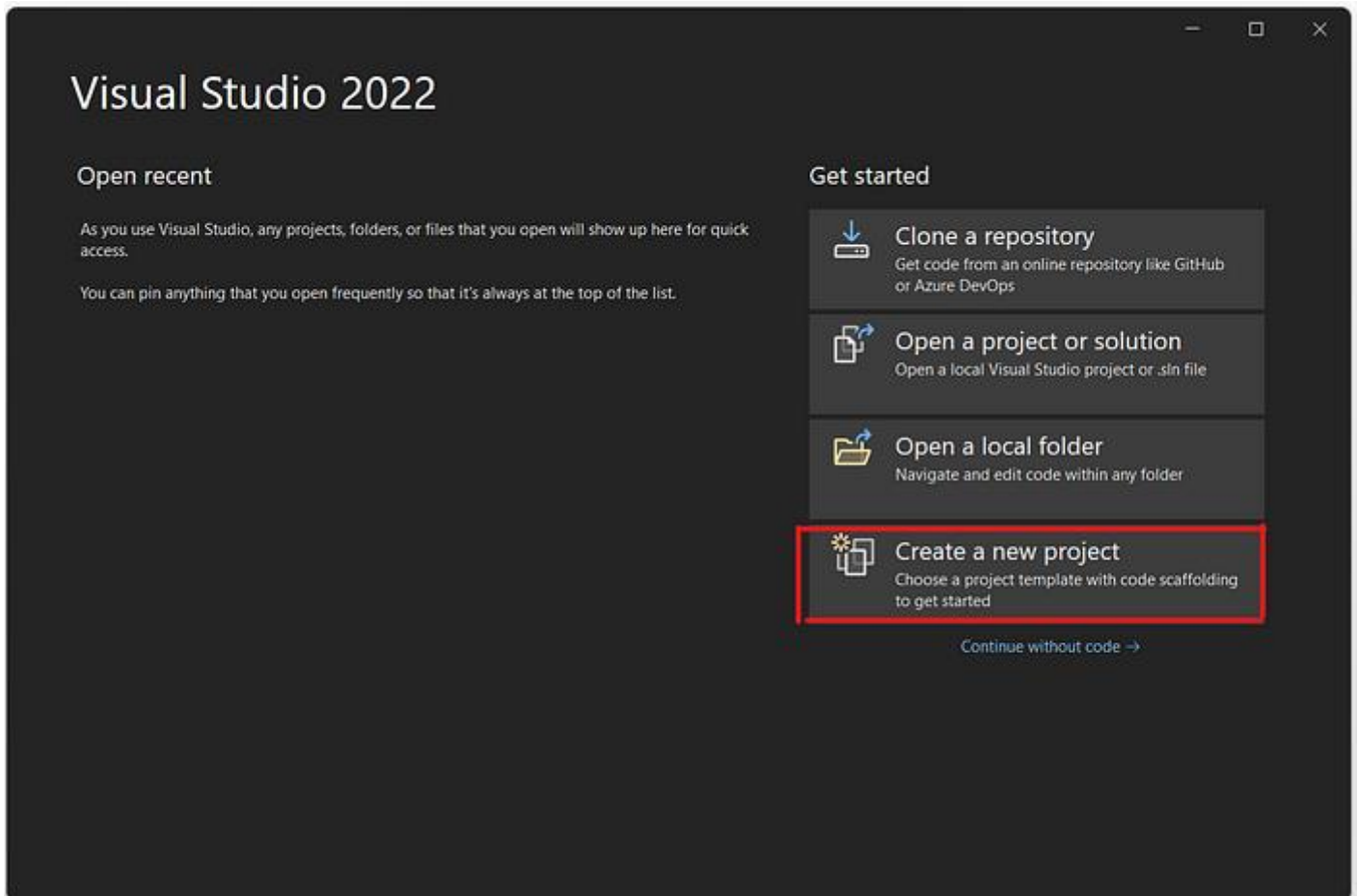
The installation of Visual Studio 2022 is now complete, and you can use it. We can observe a newly revamped Visual Studio landing page with a list of recent projects. We can also use GitHub and Azure DevOps to clone or check out the code. A project or solution, as well as a folder, can be opened. We can also start a new project.



Creating Your First Project

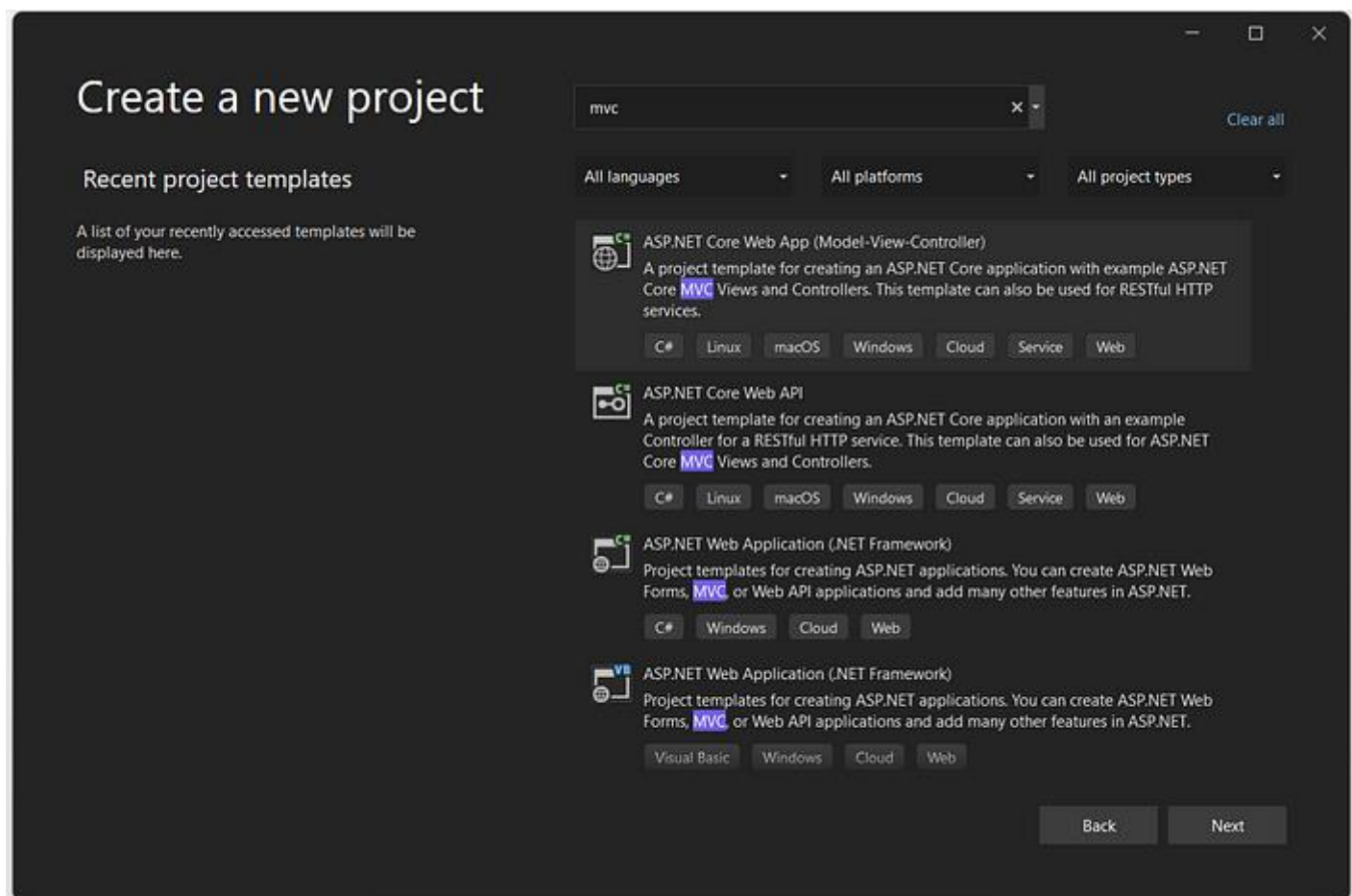
Step1

Click on Create a new Project to create a new project.

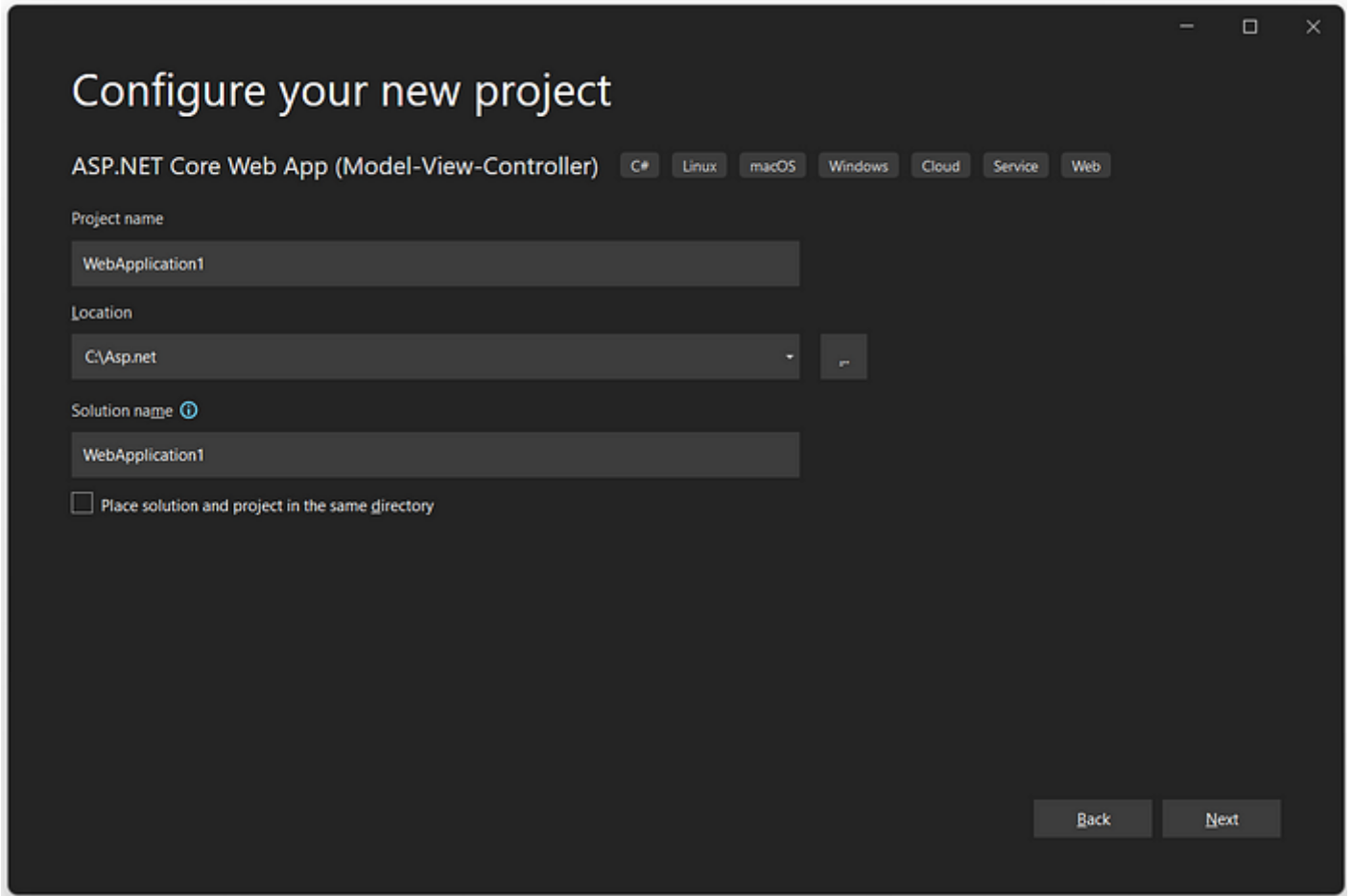


There are a variety of project types to choose from. For the time being, select the “Asp.Net Core MVC Web Application” project type.

Click Next after selecting Asp.Net Core MVC Application.



Give your project a meaningful name and select a path for it. Then click Next button.



Configure your new project

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Project name

WebApplication1

Location

C:\Asp.net

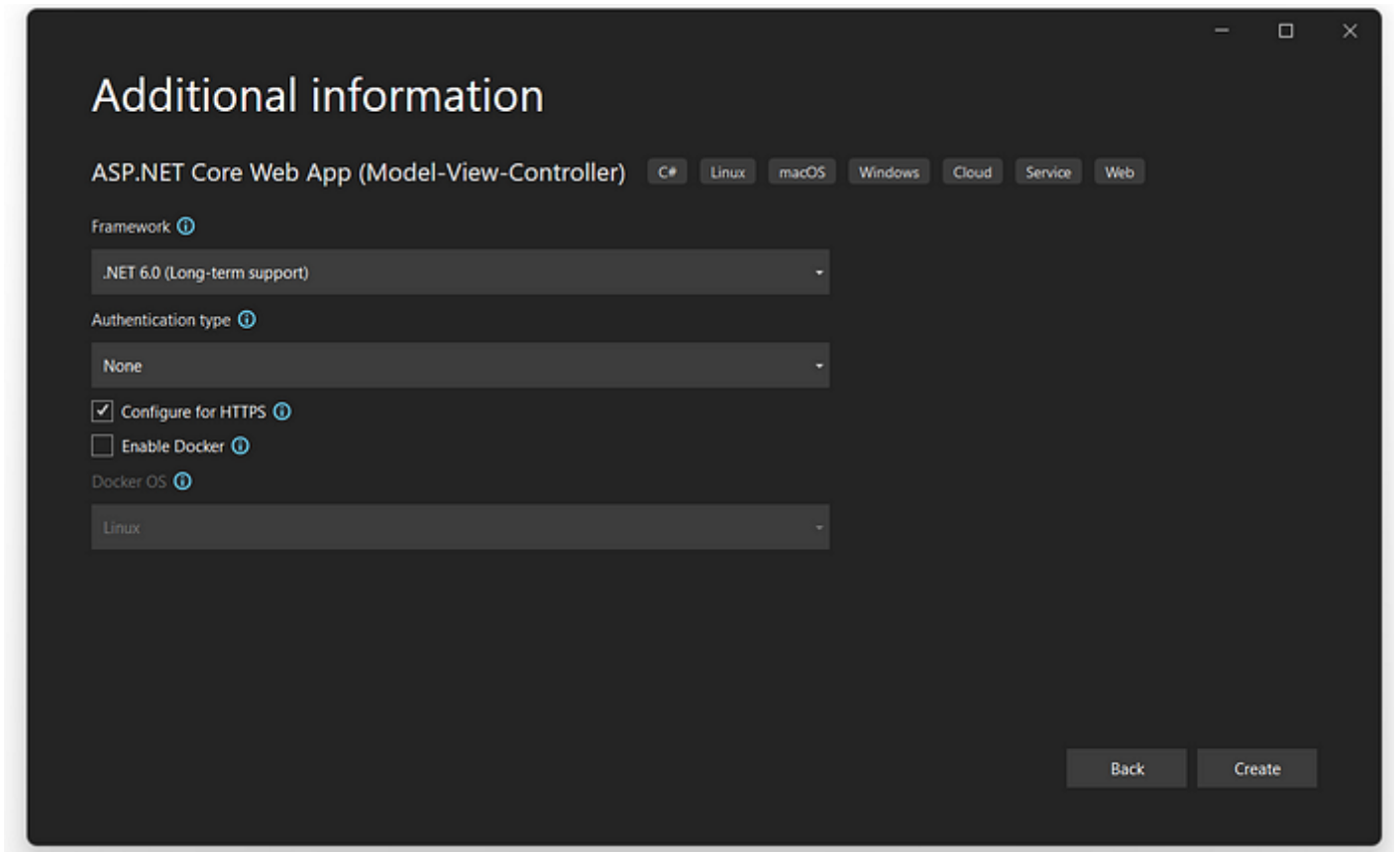
Solution name ⓘ

WebApplication1

☐ Place solution and project in the same directory

Back Next

Now, choose framework .NET 6.0. Then click the create button.



Additional information

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Framework ⓘ

.NET 6.0 (Long-term support)

Authentication type ⓘ

None

☒ Configure for HTTPS ⓘ

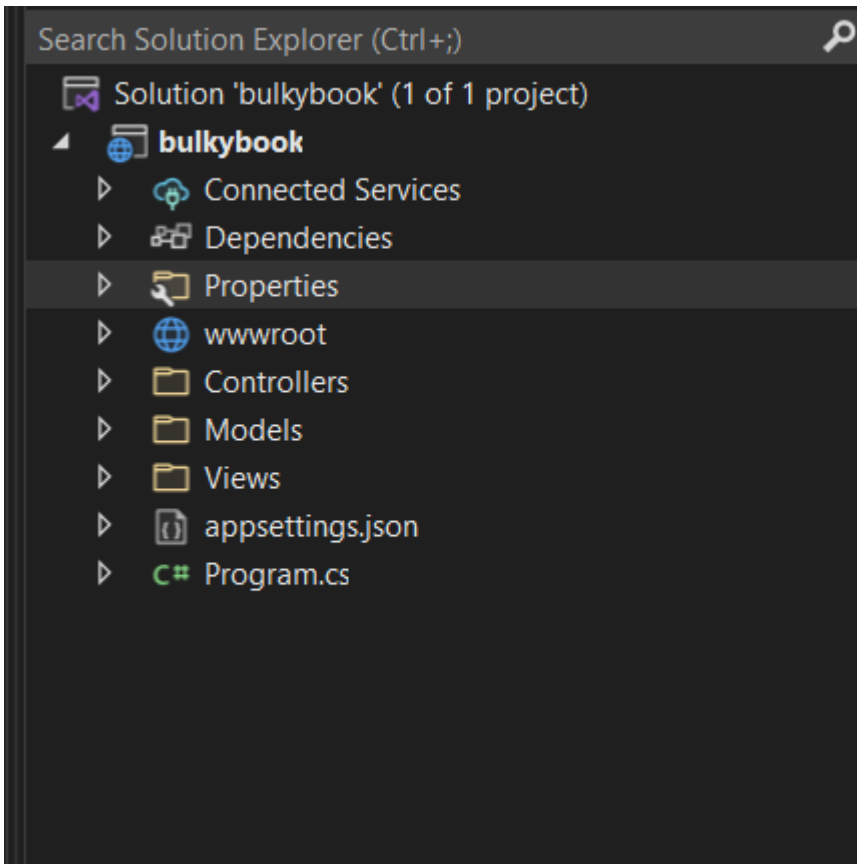
☐ Enable Docker ⓘ

Docker OS ⓘ

Linux

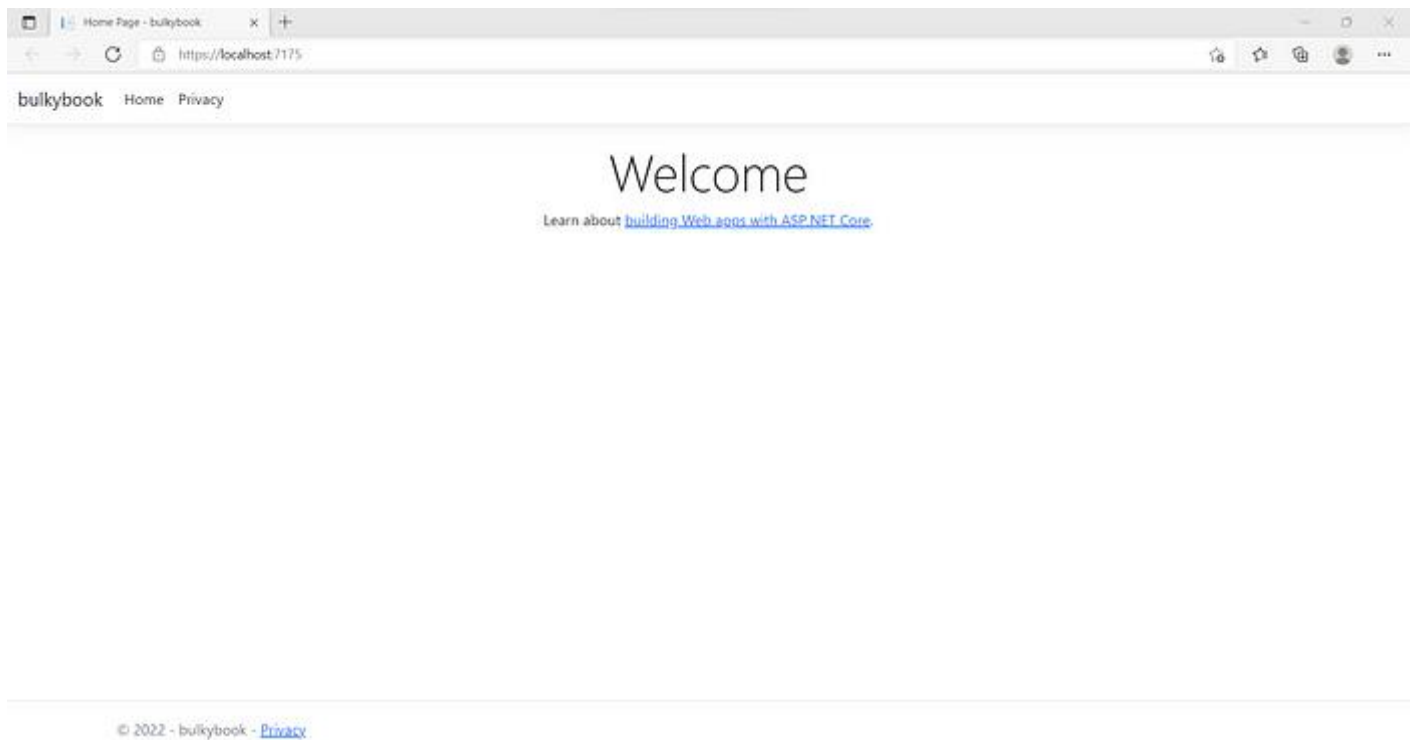
Back Create

Asp.Net Core MVC 6.0 application created and project structure is shown below,



Now, build and run the application.

OUTPUT





Installing Flutter and Configuring Android Studio for Flutter Development

Important: Path to install flutter: C:\src\flutter

Step-1: Download flutter from its official website.

Downloading the Flutter SDK also downloads the compatible version of Dart.

Filename: flutter_windows_3.3.10-stable.zip (As on 3rd January, 2023)

Step-2: Go to C: drive and create a folder named src.

Step-3: Right-click on the file mentioned in step-1, select Open with, click WinRAR.

Extract the file in C:\src folder.

The C:\src\flutter folder must list few of the following folders/files:

.git
.github
.idea
.pub-cache
bin
dev
examples
packages
.ci.yaml

etc...

Total: 25 items

Step-4 Type env in search box and click on Edit the System Environment variables.

Step-5 In the System Properties dialog box, select Advanced tab and Click on Environment Variables button shown at the bottom.

Step-6 In the Environment Variables dialog box, look at User variables for <<Username>>, double-click on Path.

Step-7 In the Edit Environment Variables dialog box, click on New, type in C:\src\flutter\bin and Click Ok.

Step-8 Go to command prompt and run then following command:

```
flutter --version
```

It should give following output:

Flutter 3.3.9 • channel stable • <https://github.com/flutter/flutter.git>

Framework • revision b8f7f1f986 (11 days ago) • 2022-11-23 06:43:51 +0900

Engine • revision 8f2221fbef

Tools • Dart 2.18.5 • DevTools 2.15.0



Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming

C:\Users\CHARUSAT>flutter doctor

Doctor summary (to see all details, run flutter doctor -v):

[V] Flutter (Channel stable, 3.3.9, on Microsoft Windows [Version 10.0.22621.674], locale en-US)

[!] Android toolchain - develop for Android devices (Android SDK version 33.0.1)

X cmdline-tools component is missing

Run `path/to/sdkmanager --install "cmdline-tools;latest"`

See <https://developer.android.com/studio/command-line> for more details.

X Android license status unknown.

Run `flutter doctor --android-licenses` to accept the SDK licenses.

See <https://flutter.dev/docs/get-started/install/windows#android-setup> for more details.

[V] Chrome - develop for the web

[X] Visual Studio - develop for Windows

X Visual Studio not installed; this is necessary for Windows development.

Download at <https://visualstudio.microsoft.com/downloads/>.

Please install the "Desktop development with C++" workload, including all of its default components

[V] Android Studio (version 2021.3)

[V] Connected device (3 available)

[V] HTTP Host Availability

! Doctor found issues in 2 categories.

Adding Flutter Plug-in to Android Studio

Step-1 Open Android Studio> File menu> Settings > Plugins (below Editor).

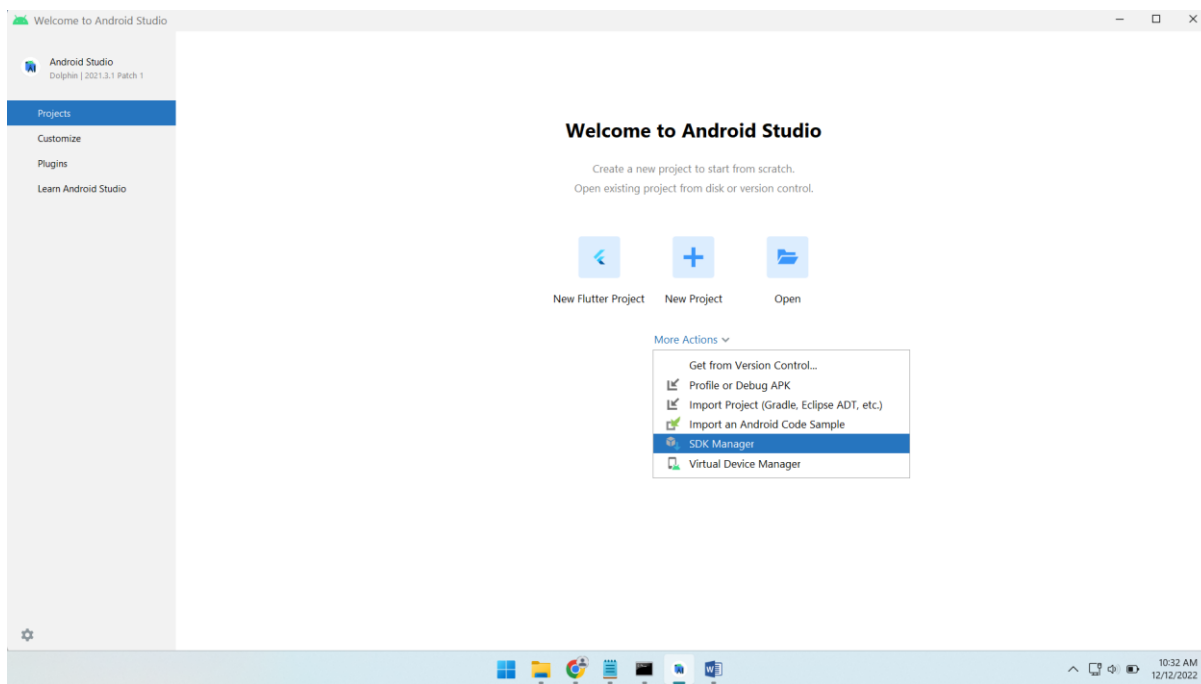
Step-2 Install: Flutter and Dart Plugins.

Configuring Android Studio for Flutter after installation

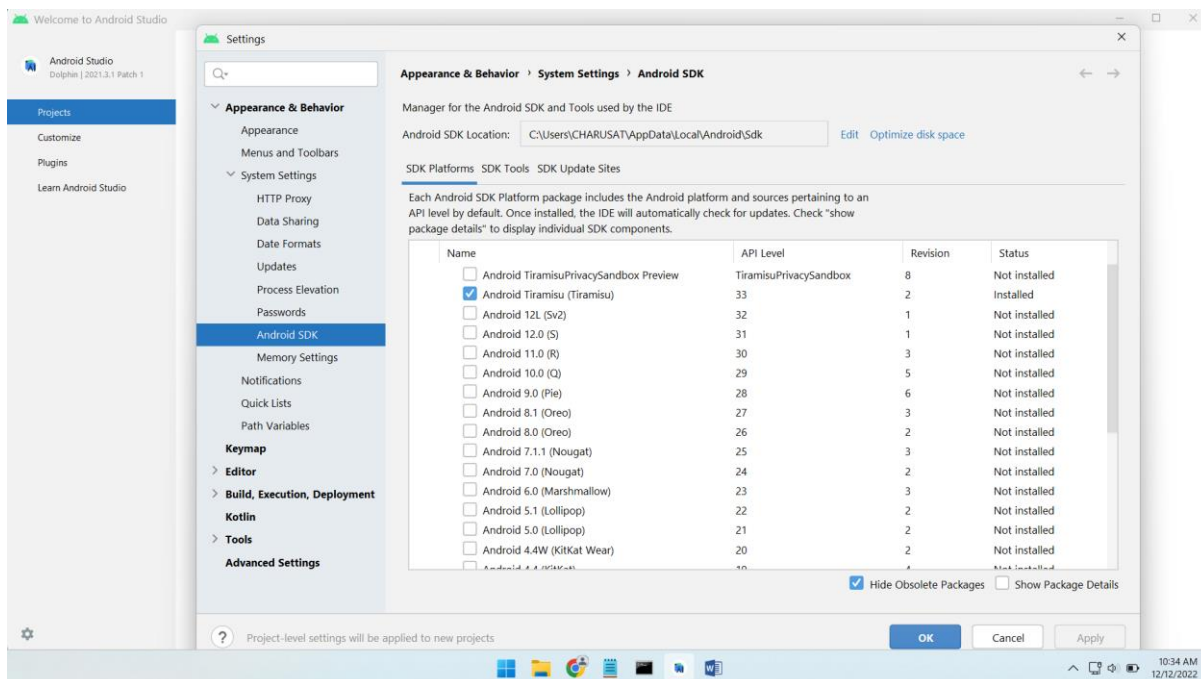
How to resolve "cmdline-tools component is missing" issue?

Step-1 Open Android Studio, click on More Actions drop-down and select SDK Manager.

Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming

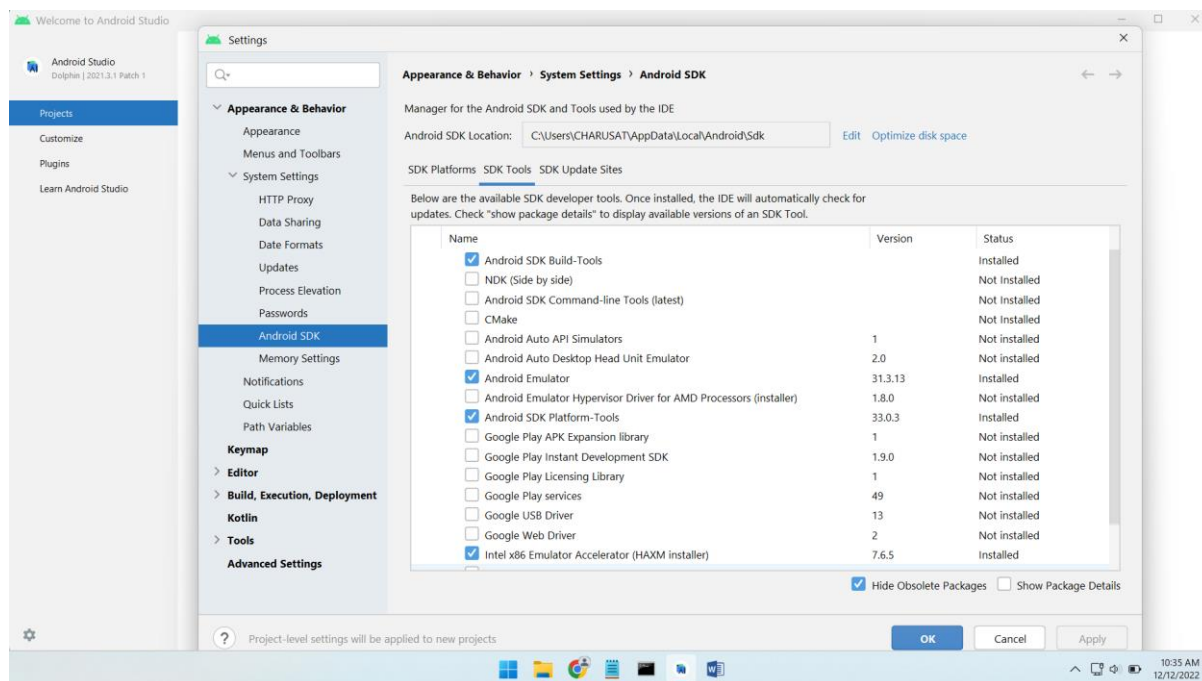


Step-2 In the Settings window, select Android SDK on LHS.

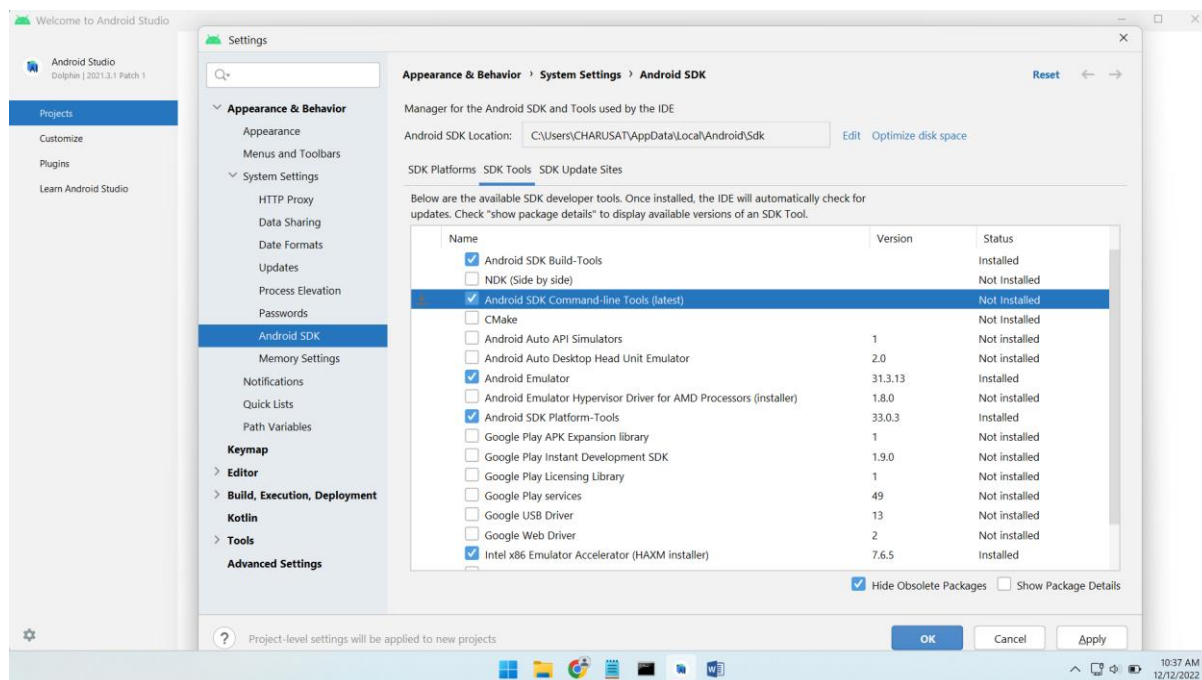


Next, click on SDK Tools tab on RHS.

Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming

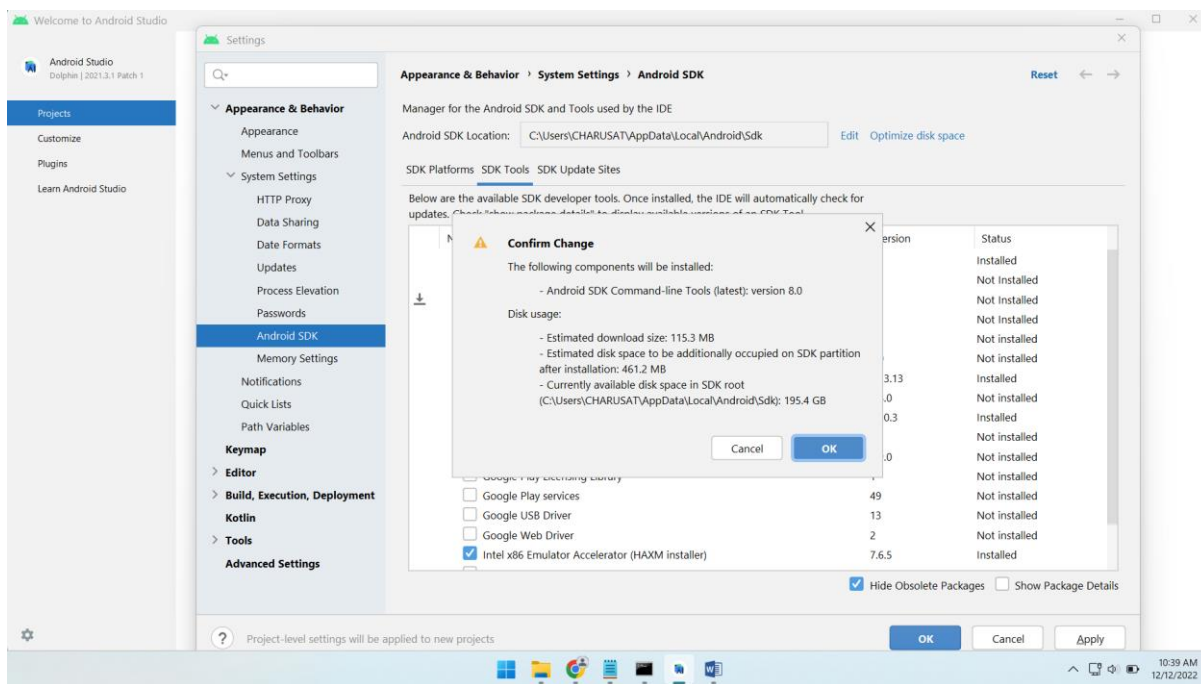


Step-3 Select Android SDK Command-line Tools(latest). Make sure you click to put the tick mark before its name. Click on Apply button.

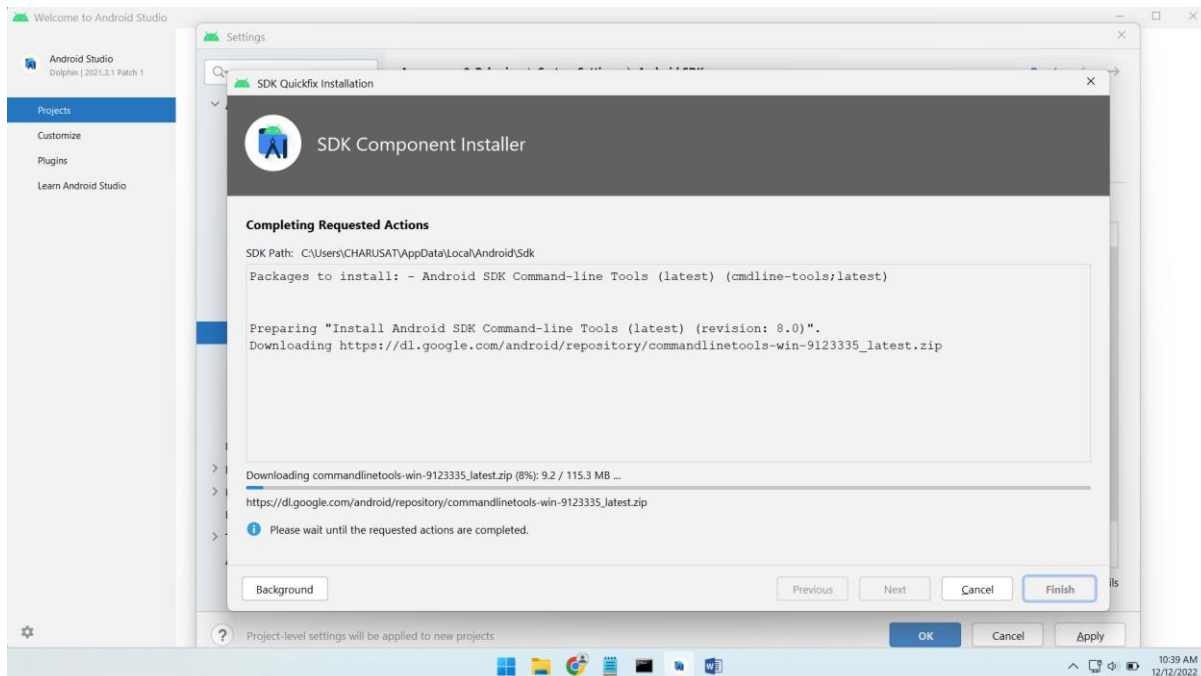


Click OK to confirm.

Tool Based Manual || MCA || SEM II || CA871 : Advanced Mobile Programming



The installation will begin.

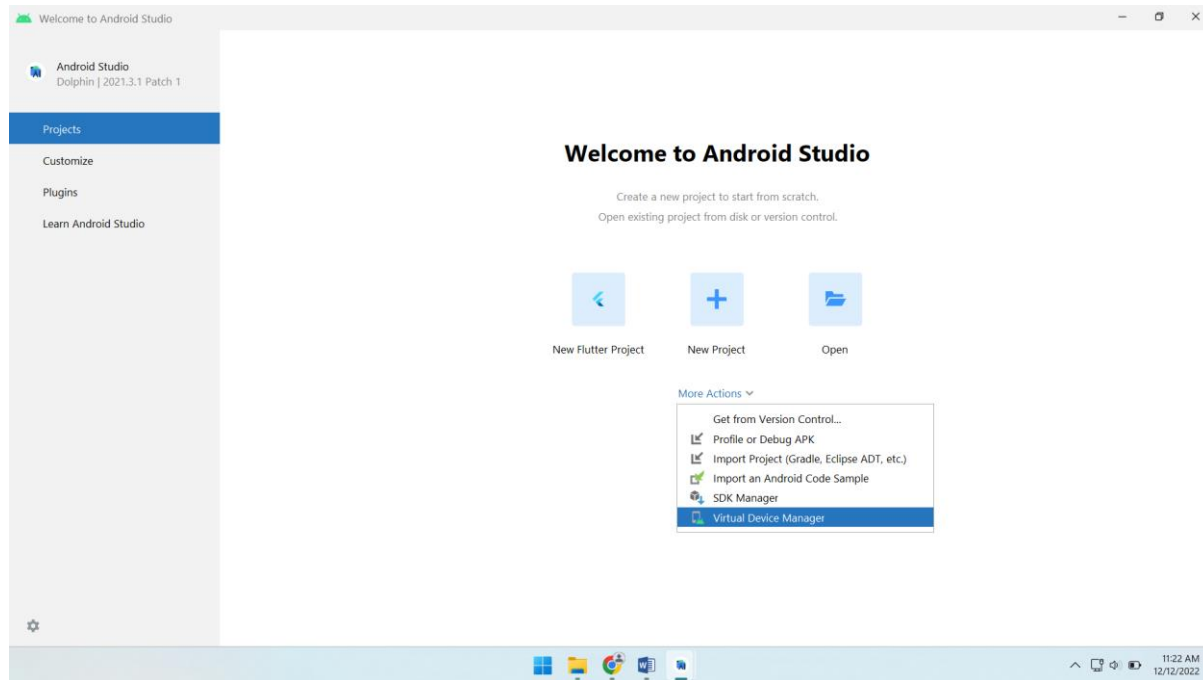


Click Finish button upon completion.

Set JAVA_HOME variable to: C:\Program Files\Java\jdk1.8.0_171 and not to C:\Program Files\Java\jdk1.8.0_171\bin

Set up the Android emulator

Step-1 Open Android Studio, click on More Actions drop-down and select Virtual Device Manager.



To change font-size in the code editor, go to File menu > Settings > Editor (on LHS) > Font > Size.



Prepared by : Dr. Kamini Solanki

Id: 1061

Contents

Preview.....2

Creating an Android Application.....3



Preview

- Android is an operating system and programming platform developed by Google for mobile phones and other mobile devices, such as tablets.
- It can run on many different devices from many different manufacturers. Android includes a software development kit (SDK) that helps you write original code and assemble software modules to create apps for Android users.
- The first beta version of the Android Software Development Kit (SDK) was released by Google in 2007 where as the first commercial version, Android 1.0, was released in September 2008.
- Now the Kotlin is official language for development instead Java.



Creating an Android Application

Tools required: Android Studio Version Dolphin

Table to Download links for tools:

Android Studio	https://developer.android.com/studio
---	---

Creating our first Android Application:

1. Open Android Studio.
2. In the Welcome to Android Studio dialog, click Start a new Android Studio project.
3. Select Empty Activity
4. Give your application a name such as My First App.
5. Make sure the Language is set to Kotlin.
6. Leave the defaults for the other fields.
7. Click Finish.